

Zaijing Li

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Research Interests

My research primarily focuses on **Embodied Agents** in both virtual and physical environments. In particular, I investigate how different forms of memory—**long-term memory**, **short-term memory**, and **parametric memory**—can be leveraged to enhance agent capabilities and performance. To this end, we have developed the **Optimus** family of agents across diverse domains.

Minecraft Agents

- **Optimus-1**: *Hybrid Multimodal Memory* for long-horizon planning and reflection [NeurIPS 2024]
- **Optimus-2**: *Action-guided Behavior Encoder* for action history sequence modeling [CVPR 2025]
- **Optimus-3**: *Dual-Router Aligned MoE* for coupling System 1 and System 2

Robotic Manipulation

- **OptimusVLA**: *Global Prior Alignment and Local Consistency Constraints* [CVPR 2026]
- **BehaviorVLA**: *Visuomotor Behavior Encoder for behavior representation learning* [ICML 2026 Oral]
- **Optimus-R**: *Inline Memory with Reusable Skills for Robotic Efficient Adaptation*

GUI Agents

- **Mirage-1**: *Hierarchical Multimodal Skills* for skill learning and updating
- **HiconAgent**: *History Context-Aware Optimization* for reinforcement learning [CVPR 2026]

Emotional Agents

- **Optimus-E**: *Short-term memory* for emotional recognition, understanding, and generation

Educations

Harbin Institute of Technology (Shenzhen)

Ph.D. advised by Prof. Liqiang Nie
Computer Science and Technology

Shenzhen, China

Sep 2023 – Present

Central South University

M.Sc. advised by Prof. Ming Zhao
Computer Science and Technology

Changsha, China

Sep 2020 – Jun 2023

Central South University

B.Sc. advised by Prof. Huihai Wang
Electronic Information Science and Technology

Changsha, China

Sep 2016 – Jun 2020

Projects & Internships

Open-World Agent

Ongoing

- **Optimus Series** [NeurIPS 2024] [CVPR 2025] [CVPR 2026]

Research Intern in Huawei Noah's Ark Lab

Nov 2024 – Apr 2025

- **Mirage-1** [Under Review]

Research Intern in Alibaba DAMO Academy

Aug 2022 – Mar 2023

- **UniSA** [ACM Multimedia 2023]

- **EmoCaps** [ACL 2022 Findings]

Selected Publications

- Bing Hu, **Zaijing Li**, Rui Shao, Junda Chen, April Hua Liu, Wei-Shi Zheng, Liqiang Nie. From Abstraction to Instantiation: Learning Behavioral Representation for Vision-Language-Action Model. *ICML 2026 Oral* (CCF-A).
- **Zaijing Li**, Bing Hu, Rui Shao, Gongwei Chen, Dongmei Jiang, Pengwei Xie, Jianye Hao, Liqiang Nie. Global Prior Meets Local Consistency: Dual-Memory Augmented Vision-Language-Action Model for Efficient Robotic Manipulation. *CVPR 2026* (CCF-A).
- **Zaijing Li**, Yuquan Xie, Rui Shao, Gongwei Chen, Dongmei Jiang, Liqiang Nie. Optimus-2: Multi-modal Minecraft Agent with Goal-Observation-Action Policy. *CVPR 2025* (CCF-A).
- **Zaijing Li**, Yuquan Xie, Rui Shao, Gongwei Chen, Dongmei Jiang, Liqiang Nie. Optimus-1: Hybrid Multimodal Memory Empowered Agents Excel in Long-Horizon Tasks. *NeurIPS 2024* (CCF-A).
- **Zaijing Li**, Meng Liu, Fengxiao Tang, Ming Zhao. UniSA: Unified Generative Framework for Sentiment Analysis. *ACM MM 2023* (CCF-A).
- **Zaijing Li**, Fengxiao Tang, Ming Zhao, and Yusen Zhu. EmoCaps: Emotion Capsule based Model for Conversational Emotion Recognition. *Findings of ACL 2022* (CCF-A).
- **Zaijing Li**, Yuquan Xie, Rui Shao, Gongwei Chen, Dongmei Jiang, Liqiang Nie. Optimus-3: Dual-Router Aligned Mixture-of-Experts Agent with Dual-Granularity Reasoning-Aware Policy Optimization. *Under Review*.

Awards

- PengCheng Laboratory Scholarship, 2025
- **Outstanding Master's Thesis Award** of Hunan Province, 2025
- CVPR 2025, Outstanding Reviewer
- CVPR 2024 - EGO4D Challenge Ego Schema Track, **1st Place Award**
- CVPR 2024 - EGO4D Challenge Natural Language Queries Track, **2nd Place Award**
- CVPR 2024 - EGO4D Challenge Goal Steps Track, **3rd Place Award**
- **Outstanding Master's Thesis Award** of Hunan Computer Federation, 2024
- **National Graduate Scholarship**, 2022
- National College Student Optoelectronic Design Competition, **Third Prize**, 2018
- Huawei Cup National College Student Intelligent Design Competition, **First Prize**, 2018

Academic Services

- **Conference Reviewer:** CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, ACM MM, ACL, AAAI.
- **Journal Reviewer:** IEEE TKDE.